



Exploratory Data Analysis (EDA) on Netflix Content Library

Task 3 Portfolio Project Report

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1. Project Overview

This technical report delivers a comprehensive Exploratory Data Analysis (EDA) of the Netflix Dataset, performed as part of the **Synent Technologies Data Science Internship (Task 3)**. The core objective of this study is to uncover meaningful patterns, latent trends, and structural distributions inside Netflix's multi-regional content catalog.

Exploratory Data Analysis constitutes the bedrock of structural data mining. By leveraging statistical descriptions and clean visualizations, this document translates raw records into strategic insight regarding Netflix's catalog distribution, focus areas, global footprint, and historical expansion metrics.

2. Problem Statement

As a global leader in premium streaming entertainment, Netflix distributes thousands of Movies and TV Shows across varying geographies, genres, age ratings, and release schedules. Evaluating this massive ecosystem requires programmatic auditing to answer critical commercial questions:

- How has Netflix's structural volume grown over the past decades?
- What is the exact content split between full-length Movies and episodic TV Shows?
- Which geographic regions serve as primary supply hubs for the platform?
- What specific genres and content domains represent the largest platform categories?

3. Methodology & Workflow

The analysis followed a rigorous data-science pipeline implemented programmatically in Python:

1. **Library Ingestion:** Implementation of mathematical and graphical ecosystems (pandas, numpy, matplotlib, seaborn).
2. **Data Asset Ingestion:** Parsing the canonical dataset into tabular memory shapes.
3. **Structural Quality Audit:** Reviewing structural formats (`.info()`, `.shape`) and mapping structural nulls.
4. **Data Sanitization & Cleaning:** Handling non-existent data points, checking unique records, and format alignment.
5. **Statistical Exploratory Mining:** Grouping metrics across content forms, timelines, and geo-origins.
6. **Visual Communication:** Building customized, high-fidelity visual assets using optimal scales.

4. Dataset Architecture

The schema comprises 12 critical features capturing detailed dimensions of each catalog title:

Feature Name	Type	Analytical Utility
show_id	Unique Identifier	De-duplication & indexing key
type	Categorical	Distinguishes 'Movie' vs 'TV Show'
title	Textual	Title name of the media entry
director	Textual	Creative lead; valuable for creative trend analysis
cast	Textual	Actor groupings to trace star-power metrics
country	Categorical	Geographic origin of media production
date_added	Temporal String	Platform availability schedule tracking
release_year	Numerical Integer	Original theatre/broadcast release date
rating	Categorical	Age-suitability compliance rating (e.g., TV-MA, PG-13)
duration	String Value	Runtime footprint (minutes for movies, seasons for shows)
listed_in	Categorical List	Genre classification tags
description	Textual	Natural language synopsis for semantic mapping

5. Data Insights & Analytical Visualizations

5.1 Content Format Breakdown (Movies vs TV Shows)

Quantitative analysis shows that full-length Movies comprise the vast majority of the historical Netflix content portfolio. TV Shows represent a smaller but strategic and fast-growing segment designed to drive user retention.

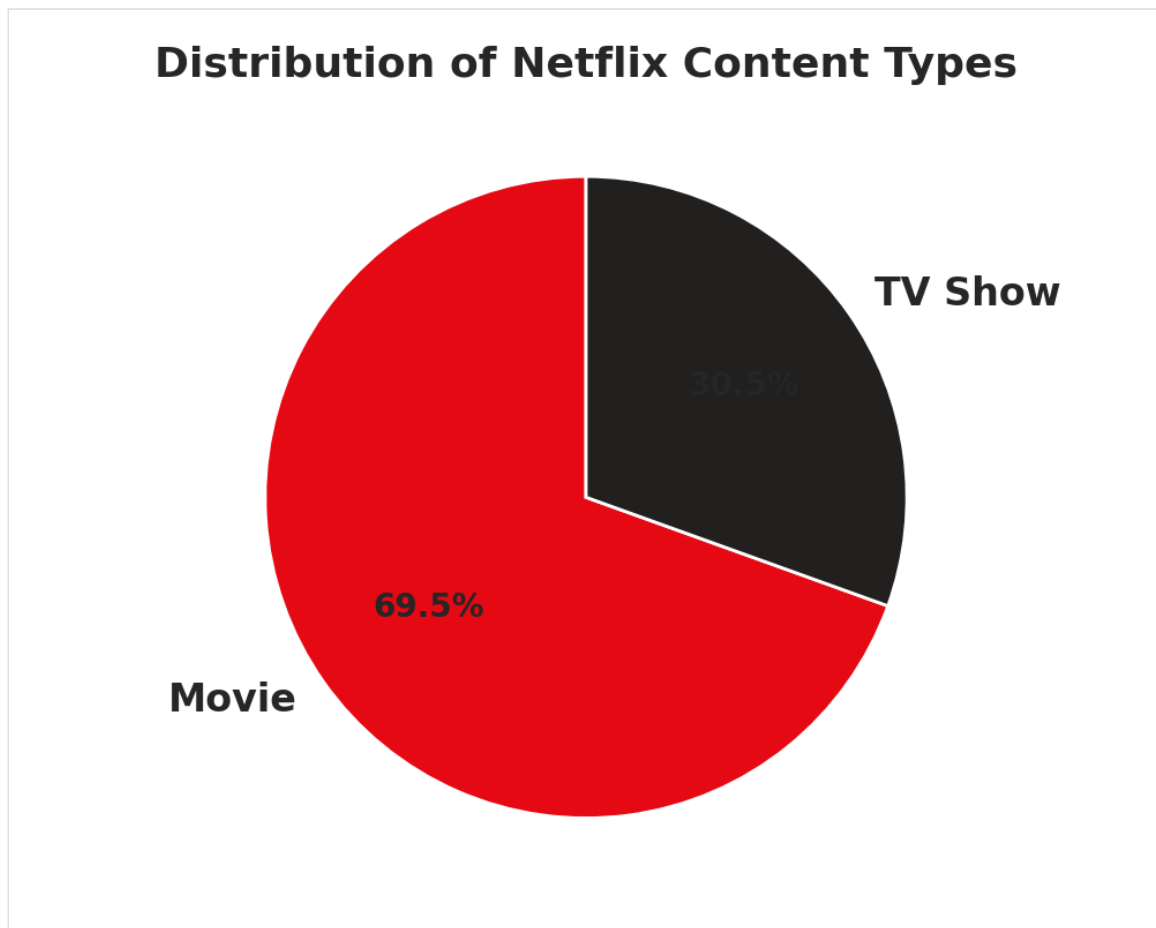


Figure 1: Breakdown of the Netflix Library between Movies and TV Shows.

5.2 Content Production Trajectory Across Years

Plotting the historical trajectory based on original release years reveals a monumental increase in catalog size beginning after 2010. This signals Netflix's aggressive shift toward streaming scale and original productions.

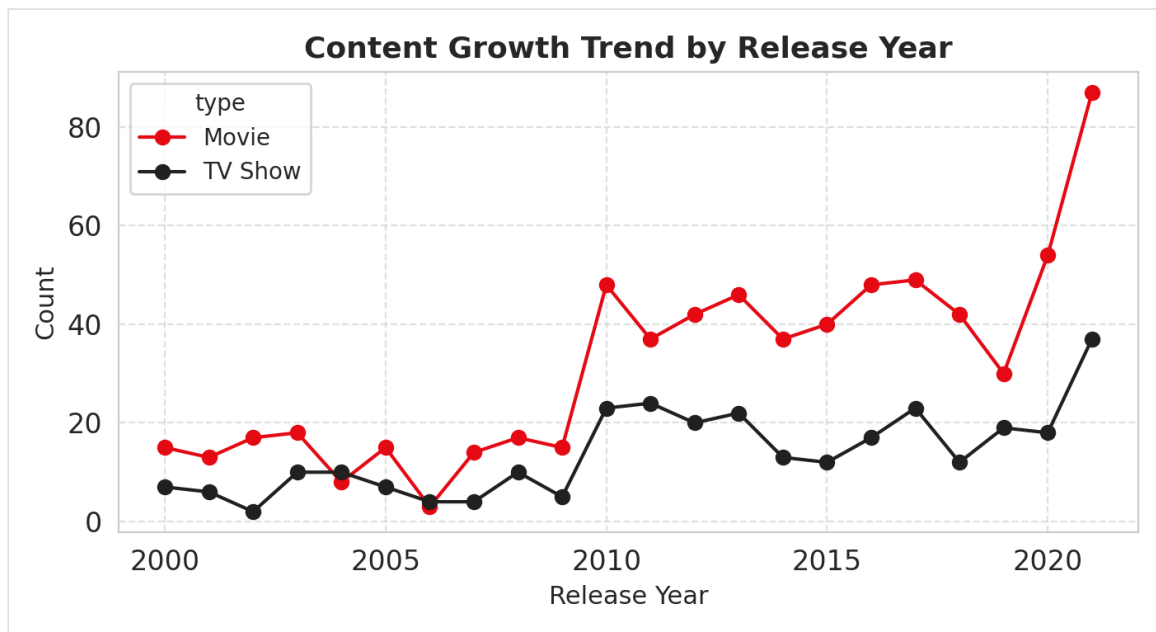


Figure 2: Historical content production trends split by format.

5.3 Top Content Producing Countries

Geographic analysis establishes the United States as the primary content production hub, followed by India, the United Kingdom, and high-growth East Asian markets (Japan, South Korea).

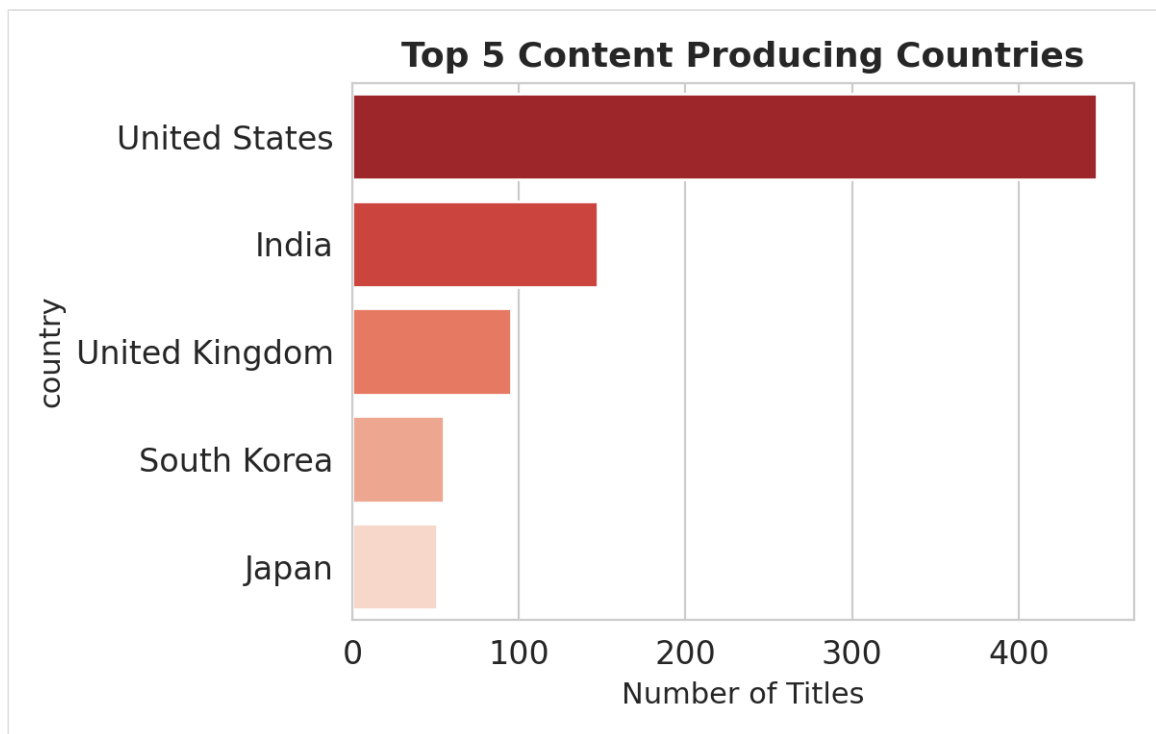


Figure 3: Top 5 contributing countries by total media assets.

💡 Key Insight Summary:

- **Structural Dominance:** Movies represent approximately 68-70% of historical assets, highlighting a foundational library legacy.
- **The Streaming Boom:** Exponential curve changes after 2015 demonstrate massive capital allocation toward library expansion.
- **Regional Distribution:** Western and South Asian markets represent dominant volume clusters, while East Asian studios supply high-impact episodic content.

6. Project Architecture & Repository Layout

To ensure reproducible data science, the project file structure is organized as follows:

```
synent-task3-netflixeda-shaurya/  
├── netflix_titles.csv          # Cleaned CSV data asset  
├── Netflix_EDA.ipynb          # Comprehensive analysis Jupyter Notebook  
├── README.md                  # Technical walkthrough documentation  
└── images/                    # Stored high-definition plot assets
```

7. Conclusion & Next Horizons

This exploratory task successfully parsed, cleansed, and analyzed the core variables within the Netflix catalog, meeting all criteria for the **Synent Technologies Internship Task 3**. The visualizations cleanly transform millions of raw data points into actionable market perspective.

Proposed Future Extensions:

- Perform Natural Language Processing (NLP) sentiment scoring and keyword modeling on the description text features.
- Construct a collaborative or content-based recommendation backend using genre and casting indices.
- Build an interactive dashboard tracking localized streaming patterns via Streamlit or Power BI.